

AHMED BARAKAT

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EDUCATION

Honours Bachelor of Artificial Intelligence

Durham College, Oshawa, ON

Sept 2024 – Apr 2028 (expected)

Bachelor of Engineering, Software Engineering

Carleton University, Ottawa, ON

Sept 2023 – Apr 2024

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, JavaScript, SQL, HTML/CSS

AI / ML: LangChain, Ollama, RAG, LLM APIs, Prompt Engineering, Hugging Face Transformers, scikit-learn, YOLOv8, OpenCV

Data & Dev Tools: Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Docker, Linux, Git, Agile (SCRUM)

Web & Frameworks: React.js, Node.js, Express.js, REST APIs, PostgreSQL, ChromaDB, Prisma, Supabase, Streamlit, Vercel, Unity

PROJECTS

AskLocal — LangChain, Ollama, ChromaDB, Streamlit

- Built a fully local, privacy-preserving RAG chatbot that answers questions about PDF documents with cited sources — no external API calls, so no data ever leaves the machine.
- Built the pipeline end to end: overlapping text chunks, local embeddings into a ChromaDB vector store, and top-k retrieval with prompts that cite file and page and answer “I don’t know” instead of hallucinating.

PulseGuard — scikit-learn, Pandas, Streamlit

- Built a predictive maintenance dashboard that forecasts equipment Remaining Useful Life from sensor data using a gradient boosting model, evaluated on machines held out entirely from training.
- Wrote a run-to-failure sensor simulator for an 80-machine fleet and engineered rolling-window features, reaching ~24-cycle MAE (~36 RMSE) on unseen machines with 120–300 cycle lifespans.

SiteWatch — YOLOv8, OpenCV, Streamlit

- Built a computer vision safety compliance monitor that detects people in a photo and flags missing hard hats or high-visibility vests, combining a pretrained object detector with color-based analysis.
- Debugged real-world failures (missed hard hats in dim lighting, vest colors bleeding between people standing close together) by widening HSV brightness ranges and excluding pixels shared by overlapping detections.

FitRift — Node.js, Express, React, PostgreSQL, Prisma, Supabase, Groq LLM

- Built a full-stack fitness RPG website that uses an LLM to parse natural-language workout logs, tracks XP, personal records, and rank progression against the user's own baseline, with real authentication and a live deployed demo.
- Designed an Express REST API with Prisma, Supabase JWT auth, rate limiting, and LLM JSON-mode parsing that turns phrases like “20 pushups, felt tough” into structured sets, reps, and effort ratings.

PROFESSIONAL EXPERIENCE

Construction Laborer, LIUNA Union (seasonal)

Mulrooney Company, Kingston, ON

May – Jul 2025, May – Jul 2026

- Performed general labor on active construction sites — material handling, site preparation, and equipment support — in accordance with LIUNA safety standards and protocols.
- Collaborated with skilled tradespeople and site supervisors across two consecutive summer seasons to complete tasks efficiently while maintaining a safe, organized work site.

Sales Associate (part-time)

Shoppers Drug Mart, Kingston, ON

Jul 2022 – Sep 2023

- Implemented effective communication and interpersonal abilities to promptly address all customer requirements while upholding a tidy and orderly store environment.
- Operated efficiently in a fast-paced setting, serving a minimum of 100 customers per shift during holiday periods.